

Draw HI parallel to the tangent at P , meeting the diameter EC produced. The great focal property of the ellipse is that $PS + PH = 2AC$: the sum of the two focal radii is constant and equals the major axis. From this Newton deduces the pivotal equality $EP = AC$, the semi-major axis itself. This single relation binds the ellipse's focal geometry to the measurable lengths the force calculation will demand.